

Lesson 6-1: Division Expressions with Fractions and Whole Numbers

Connect & apply · 8 tasks · Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

4.8 ÷ 2? How many fourths are in 3? Reciprocal of 5?

TASK 6 • QUICK CHECK B

6. $4 \div 2$? How many thirds are in 4? Reciprocal of 8?

TASK 7 • QUICK CHECK C

9.6 ÷ 4? How many halves are in 5? Reciprocal of 1/6?

TASK 8 · TRANSFER & EXTEND

A ribbon is 6 feet long. How many $\frac{3}{4}$ -foot pieces can be cut from it?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Write $2\frac{3}{4}$ as an improper fraction.

TASK 2 • FOUNDATION 2

Compute $\frac{2}{3} \times \frac{3}{5}$.

TASK 3 • FOUNDATION 3

Find the reciprocal of $\frac{4}{7}$.

TASK 4 • FOUNDATION 4

Simplify $\frac{18}{24}$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-2: Division Expressions with Fractions and Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

2 3/4 as an improper fraction? Reciprocal of 3/5?
2/3 × 5/4?

TASK 6 • QUICK CHECK B

3 2/5 as an improper fraction? Reciprocal of 4/7?
3/4 × 2/5?

TASK 7 • QUICK CHECK C

1 5/6 as an improper fraction? Reciprocal of 9/2?
5/6 × 3/10?

TASK 8 • TRANSFER & EXTEND

How many 1 1/2-cup servings are in 6 cups?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-3: Explore Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $3 \times 3 \times 3 \times 3$.

TASK 2 · FOUNDATION 2

Evaluate 5^2 .

TASK 3 • FOUNDATION 3

Evaluate $10 \times 10 \times 10$.

TASK 4 • FOUNDATION 4

How many factors of 2 appear in $2 \times 2 \times 2 \times 2 \times 2$?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-3: Explore Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

6²? Write $2 \times 2 \times 2 \times 2$ as a power. 10³?

TASK 6 · QUICK CHECK B

7²? Write $3 \times 3 \times 3 \times 3 \times 3$ as a power. 10⁴?

TASK 7 · QUICK CHECK C

4³? Write $5 \times 5 \times 5$ as a power. 10²?

TASK 8 · TRANSFER & EXTEND

Which is greater, 3⁴ or 4³?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2.c

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Evaluate 4^3 .

TASK 2 • FOUNDATION 2

Evaluate $2 + 3^2 \times 4$.

TASK 3 • FOUNDATION 3

Evaluate $(8 + 4)/3$.

TASK 4 • FOUNDATION 4

Write an expression for 7 less than the product of 3 and 5.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-4: Write and Evaluate Numerical Expressions with Exponents

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.2.c

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

$4 + 6 \div 2$? $3 + 2^2 \times 5$? Which operation comes first?

TASK 6 · QUICK CHECK B

$8 + 12 \div 4$? $5 + 3^2 \times 2$? Which operation comes first?

TASK 7 · QUICK CHECK C

$20 - 6 \times 2$? $4 + 2^3 \div 4$? Which operation comes first?

TASK 8 · TRANSFER & EXTEND

Evaluate $2 \times (3 + 4)^2 - 10$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Evaluate $n + 8$ when $n = 6$.

TASK 2 · FOUNDATION 2

Evaluate $2x + 3$ when $x = 5$.

TASK 3 · FOUNDATION 3

Evaluate $4n$ when $n = 7$.

TASK 4 · FOUNDATION 4

Write an expression for 9 more than a number n .

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-5: Write and Evaluate Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.2

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

Evaluate $4n + 3$ when $n = 5$. Write an expression for "7 more than a number."

TASK 6 • QUICK CHECK B

Evaluate $3n + 8$ when $n = 6$. Write an expression for "5 less than a number."

TASK 7 • QUICK CHECK C

Evaluate $2n - 7$ when $n = 10$. Write an expression for "the product of 6 and a number."

TASK 8 • TRANSFER & EXTEND

Write an expression for the cost of n tickets at \$12 each plus a \$5 fee, then find the cost for 4 tickets.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Simplify $5x + 2x$.

TASK 2 · FOUNDATION 2

Evaluate $3x + 4$ when $x = 2$.

TASK 3 • FOUNDATION 3

Evaluate $3(4 + 5)$ using distribution.

TASK 4 • FOUNDATION 4

Compare $2(x + 3)$ and $2x + 6$ at $x = 4$. Give both values.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-6: Identify Equivalent Algebraic Expressions

Connect & apply • 8 tasks • Prepares for Standard 6.EE.A.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

$5x + 2x = ?$ Evaluate $2(x + 3)$ and $2x + 6$ at $x = 4$
— are they the same?

TASK 6 • QUICK CHECK B

$8y + 3y = ?$ Evaluate $3(x + 2)$ and $3x + 6$ at $x = 5$
— are they the same?

TASK 7 • QUICK CHECK C

$9a - 4a = ?$ Evaluate $4(x + 1)$ and $4x + 4$ at $x = 3$
— are they the same?

TASK 8 • TRANSFER & EXTEND

Are $3(2x + 4)$ and $6x + 7$ equivalent? Justify.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $84 \div 7$.

TASK 2 • FOUNDATION 2

List every positive factor of 18.

TASK 3 • FOUNDATION 3

List the first four positive multiples of 7.

TASK 4 • FOUNDATION 4

Which of 2, 3, 5, 9, and 10 divide 135 evenly?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-7: Find Factors and Multiples

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

All the factors of 18? First five multiples of 6?
GCF of 18 and 24?

TASK 6 • QUICK CHECK B

All the factors of 24? First five multiples of 8?
GCF of 24 and 36?

TASK 7 • QUICK CHECK C

All the factors of 20? First five multiples of 5?
GCF of 20 and 30?

TASK 8 • TRANSFER & EXTEND

You have 24 pencils and 36 erasers to split into identical kits with nothing left over. What is the largest number of kits?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-8: Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Find the greatest common factor of 18 and 30.

TASK 2 · FOUNDATION 2

Factor $12x + 20$ using the greatest common factor.

TASK 3 • FOUNDATION 3

Simplify $4a + 3a + 2$.

TASK 4 · FOUNDATION 4

Compute $6(10 + 4)$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-8: Generate Equivalent Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

GCF of 18 and 24? Expand $3(x + 5)$. Factor $4x + 12$.

TASK 6 · QUICK CHECK B

GCF of 20 and 30? Expand $5(x + 4)$. Factor $6x + 18$.

TASK 7 · QUICK CHECK C

GCF of 12 and 16? Expand $4(x + 7)$. Factor $9x + 15$.

TASK 8 · TRANSFER & EXTEND

Factor $24x + 36$ completely, then check by expanding.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Connect & apply · 8 tasks · Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

TASK 1 • FOUNDATION 1

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Connect & apply · 8 tasks · Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

TASK 5 • QUICK CHECK A

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Connect & apply · 8 tasks · Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

TASK 1 • FOUNDATION 1

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-10: Divide Mixed Numbers

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A
Write $3\frac{1}{2}$ as an improper fraction. $7/2 \times 4/1$?
Write $28/2$ as a whole number.

TASK 6 • QUICK CHECK B
Write $2\frac{1}{4}$ as an improper fraction. $9/4 \times 4/3$?
Write $36/12$ as a whole number.

TASK 7 • QUICK CHECK C
Write $4\frac{2}{3}$ as an improper fraction. $14/3 \times 3/2$?
Write $42/6$ as a whole number.

TASK 8 • TRANSFER & EXTEND
A painter has $6\frac{2}{3}$ quarts of paint. Each door needs $1\frac{1}{3}$ quarts. How many doors can be painted? Check by multiplying back.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-11: Fraction Division Problem Solving

Connect & apply • 8 tasks • Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

A jug holds $\frac{7}{8}$ quart. Each cup holds $\frac{1}{8}$ quart.
Write the division equation and name the dividend and the divisor.

TASK 3 • FOUNDATION 3

Compute $6 \times \frac{1}{8}$.

TASK 2 • FOUNDATION 2

Compute $\frac{5}{6} \div \frac{1}{3}$.

TASK 4 • FOUNDATION 4

Without computing, is $\frac{3}{4} \div \frac{1}{4}$ greater than or less than $\frac{3}{4}$? Explain.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-11: Fraction Division Problem Solving

Connect & apply · 8 tasks · Prepares for Standard 6.NS.A.1

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A
In 'How many $\frac{1}{8}$ -pound casts can be made from $\frac{7}{8}$ pound?', which number is the dividend? $\frac{7}{8} \div \frac{1}{8}$? $7 \times \frac{1}{8}$?

TASK 6 · QUICK CHECK B
In 'How many $\frac{1}{6}$ -mile checkpoints fit on a $\frac{5}{6}$ -mile trail?', which number is the dividend? $\frac{5}{6} \div \frac{1}{6}$? $5 \times \frac{1}{6}$?

TASK 7 · QUICK CHECK C
In 'How many $\frac{3}{10}$ -liter vials can be filled from $\frac{9}{10}$ liter?', which number is the dividend? $\frac{9}{10} \div \frac{3}{10}$? $3 \times \frac{3}{10}$?

TASK 8 · TRANSFER & EXTEND
A $\frac{3}{4}$ -pound bag of clay is shared equally by 3 students. A second problem asks how many $\frac{3}{4}$ -pound bags make 3 pounds. Write and solve both equations and explain why the answers differ.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-12: Least Common Multiple

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

What is the 7th multiple of 8?

TASK 2 • FOUNDATION 2

Compute 9×12 .

TASK 3 • FOUNDATION 3

Does 6 divide 45 evenly? Explain how you know.

TASK 4 • FOUNDATION 4

Write 24 as a product of prime factors.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-12: Least Common Multiple

Connect & apply • 8 tasks • Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 • QUICK CHECK A

List the first 5 multiples of 6. 8×7 ? Does 4 divide 30 evenly?

TASK 6 • QUICK CHECK B

List the first 5 multiples of 8. 9×6 ? Does 3 divide 42 evenly?

TASK 7 • QUICK CHECK C

List the first 5 multiples of 12. 7×12 ? Does 6 divide 40 evenly?

TASK 8 • TRANSFER & EXTEND

Cups come in stacks of 6 and lids in packs of 8. What is the smallest number of cups and lids that match exactly, and how many stacks and packs is that?

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-13: Prime Factorization

Connect & apply · 8 tasks · Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Compute 7×9 .

TASK 2 · FOUNDATION 2

Is 51 prime or composite? Give a factor pair as evidence.

TASK 3 · FOUNDATION 3

135 ends in 5, so 5 is a factor. Compute $135 \div 5$.

TASK 4 · FOUNDATION 4

Evaluate $2^3 \times 3$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-13: Prime Factorization

Connect & apply · 8 tasks · Prepares for Standard 6.NS.B.4

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A
Is 39 prime or composite? 6×7 ? Evaluate 2^3 .

TASK 6 · QUICK CHECK B
Is 57 prime or composite? 8×9 ? Evaluate 3^2 .

TASK 7 · QUICK CHECK C
Is 43 prime or composite? 7×12 ? Evaluate 5^2 .

TASK 8 · TRANSFER & EXTEND
Find the smallest whole number whose prime factorization uses 2, 3 and 5, each exactly twice.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-14: The Distributive Property

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 • FOUNDATION 1

Compute $8 \times 40 + 8 \times 3$.

TASK 2 · FOUNDATION 2

What does $4n$ mean? Write it as a product and describe it in words.

TASK 3 • FOUNDATION 3

Find the greatest common factor of 24 and 36.

TASK 4 · FOUNDATION 4

A rectangle is 6 units tall. Its width is split into parts of 10 and 4 units. Add the two part areas to find the total area.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-14: The Distributive Property

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A

$9 \times 30 + 9 \times 4$? GCF of 12 and 18? What does 5y mean?

TASK 6 · QUICK CHECK B

$7 \times 50 + 7 \times 3$? GCF of 16 and 24? What does 8m mean?

TASK 7 · QUICK CHECK C

$6 \times 70 + 6 \times 2$? GCF of 15 and 25? What does 2k mean?

TASK 8 · TRANSFER & EXTEND

Write $42 + 56$ as a product using the greatest common factor and the distributive property.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-15: Simplify Algebraic Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 1 · FOUNDATION 1

Name the coefficient and the constant in $9k + 4$.

TASK 2 · FOUNDATION 2

In the term p , what is the coefficient?

TASK 3 · FOUNDATION 3

Expand $3(2w + 5)$.

TASK 4 · FOUNDATION 4

Evaluate $4x + 7$ when $x = 3$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner

Lesson 6-15: Simplify Algebraic Expressions

Connect & apply · 8 tasks · Prepares for Standard 6.EE.A.3

Name: _____ Date: _____ Period: _____

Show complete thinking. Label units. Use an equation, sketch, or model. Accuracy before speed.

TASK 5 · QUICK CHECK A
Coefficient of y in $y + 8$? Expand $2(3a + 4)$.
Evaluate $5x + 1$ when $x = 4$.

TASK 6 · QUICK CHECK B
Coefficient of m in $m - 3$? Expand $4(2b + 1)$.
Evaluate $3x + 2$ when $x = 5$.

TASK 7 · QUICK CHECK C
Coefficient of k in $7 + k$? Expand $5(c + 3)$.
Evaluate $6x + 4$ when $x = 2$.

TASK 8 · TRANSFER & EXTEND
Simplify $3(2p + 4) + 2(p + 1)$, then evaluate when $p = 3$.

One strategy or model I used: _____

My next step: ☐ Explain my steps ☐ Try another problem ☐ Review with a partner